COMP 2121 DISCRETE MATHEMATICS | LAB 7

Exercise 1: A sequence of numbers is defined as:

$\boxed{\text{note } a_3 \neq a3 \neq a^3 \,!}$

$a_1 = 2$

$a_2 = 5$

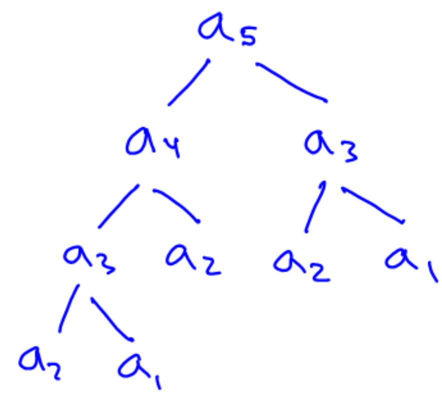
$a_n = a_{n-1} + a_{n-2}, \quad n \geq 3$

Determine a_5 and discuss recursion tree.

$a_5 = a_4 + a_3 = 12 + 7 = 19$

$a_4 = a_3 + a_2 = 7 + 5 = 12$

$a_3 = a_2 + a_1 = 2 + 5 = 7$



Exercise 2: Give a recursive definition for each of the following sequences:

a) $\boxed{a_n = 5n,}$ $n \geq 1$

$\boxed{\begin{array}{l} a_1 = 5 \\ a_n = a_{n-1} + 5;\ n \geq 2 \end{array}}$

$a_1 = 5(1) = 5$

$a_{n-1} = 5(n-1)$

$a_{n-1} = 5n - 5$

$a_{n-1} = a_n - 5$

$a_n = a_{n-1} + 5$

c) $\boxed{a_n = 2^n + 1,}$ $n \geq 1$

$\boxed{\begin{array}{l} a_1 = 3 \\ a_n = 2a_{n-1} - 1;\ n \geq 2 \end{array}}$

$a_1 = 2^1 + 1 = 3$

$a_{n-1} = 2^{n-1} + 1$

$a_{n-1} = \dfrac{2^n}{2} + 1$

$a_{n-1} = \dfrac{a_n - 1}{2} + 1$

$2a_{n-1} = a_n - 1 + 2$

$2a_{n-1} = a_n + 1$

$a_n = 2a_{n-1} - 1$

Compare the formulas:

n	$a_n = 2^n + 1$	$a_1 = 3$, $a_n = 2a_{n-1} - 1$
1	$a_1 = 2^1 + 1 = 3$	$a_1 = 3$
2	$a_2 = 2^2 + 1 = 5$	$a_2 = 2(3) - 1 = 5$
3	$a_3 = 2^3 + 1 = 9$	$a_3 = 2(5) - 1 = 9$
4	$a_4 = 2^4 + 1 = 17$	$a_4 = 2(9) - 1 = 17$

b) $\boxed{a_n = 3n + 8,}$ $n \geq 1$

$\boxed{\begin{array}{l} a_1 = 11 \\ a_n = a_{n-1} + 3;\ n \geq 2 \end{array}}$

$a_1 = 3(1) + 8 = 11$

$a_{n-1} = 3(n-1) + 8$

$a_{n-1} = 3n - 3 + 8$

$a_{n-1} = a_n - 3$

$a_n = a_{n-1} + 3$

d) $\boxed{a_n = 6,}$ $n \geq 1$

$a_1 = 6$

$a_{n-1} = 6$

$a_{n-1} = a_n$

$a_n = a_{n-1}$

$\boxed{\begin{array}{l} a_1 = 6 \\ a_{n-1} = a_n;\ n \geq 2 \end{array}}$

write in ordinary notation

Exercise 3: ~~Determine the value of the following summations:~~

a) $\sum_{k=2}^{5} k^2 + 2 + 3 = 2^2 + 3^2 + 4^2 + 5^2 + 2 + 3$

upper - lower + 1 = 3 - (-2) + 1 = 6

b) $\sum_{j=-2}^{3} (j^3 + 2) + 3 = ((-2)^3 + 2) + ((-1)^3 + 2) + (0^3 + 2) + (1^3 + 2) + (2^3 + 2) + (3^3 + 2) + 3$

Exercise 4: Express the following using the summation notation:

a) $1+2+3+4+5+6 = \sum_{i=1}^{6} i$

b) $5+6+7+8 = \sum_{i=5}^{8} i$

c) $2+2+2+2+2 = \sum_{i=1}^{5} 2$

no j

$$\boxed{\sum_{j=a}^{b} \text{const} = (b-a+1) * \text{const}}$$

d) $\frac{2}{3!}+\frac{2}{4!}+\frac{2}{5!}+\cdots+\frac{2}{(n+2)!} = \sum_{i=1}^{n} \frac{2}{(i+2)!}$

Exercise 5: Given $\sum_{i=0}^{10}(2a_1 + 3a_2) = 209$ and $a_1 = 2$, find a_2.

no i ⇒ const!

$$\sum_{i=0}^{10} (4 + 3a_2) = 209$$

$$(10 - 0 + 1)(4 + 3a_2) = 209$$

$$4 + 3a_2 = \frac{209}{11}$$

$$3a_2 = 19 - 4$$

$$a_2 = \frac{15}{3}$$

$$a_2 = 5$$

$a_1 + a_2 + \dots + a_{100} = 500$

Exercise 6: Given $\sum_{j=1}^{100} a_j = 500$ and $\sum_{k=0}^{49} a_{k+1} + \sum_{i=48}^{98} a_{i+2} + a_{50} = 506$, find a_{50}.

Pay attention to the variable

NOTE

$\sum_{i=1}^{100} a_j = 500 \Rightarrow a_j = 5$ ✓

$\sum_{j=1}^{100} a_j = 500 \cancel{\Rightarrow} a_j = 5$! ✗

$$\sum_{i=1}^{50} a_i + \sum_{i=50}^{100} a_i + a_{50} = 506$$

$$\sum_{i=1}^{50} a_i + \underbrace{a_{50} + \sum_{i=51}^{100} a_i}_{} + a_{50} = 506$$

$$\sum_{i=1}^{100} a_i + a_{50} + a_{50} = 506$$

$$500 + 2a_{50} = 506 \Rightarrow \underline{a_{50} = 3}$$

Exercise 7: Use formula $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$ to find:

must start at 1 !

a) Sum $10 + 11 + 12 + \dots + n + (n+1)$ answer: $\frac{n^2 + 3n - 88}{2}$

$$\sum_{i=10}^{n+1} i = \sum_{i=1}^{n+1} i - \sum_{i=1}^{9} i$$

$$= \frac{(n+1)((n+1)+1)}{2} - \frac{9(9+1)}{2}$$

$$= \frac{(n+1)(n+2)}{2} - \frac{9(10)}{2} = \frac{n^2+2n+n+2-90}{2}$$

$$= \frac{n^2+3n-88}{2}$$

b) The value of `counter` after the following programming segment is executed

```
counter = 10;
for (j = 6; j <= 100; j++)
        counter = counter + 2*(j + 4);
```

include brackets

answer: `counter` = 10,840

$$counter = 10 + \sum_{j=6}^{100} 2(j+4) = 10 + 2\sum_{j=6}^{100}(j+4) = 10 + 2\left(\sum_{j=6}^{100} j + \sum_{j=6}^{100} 4\right)$$

variable, need to fix j=6

const, no need to fix j=6

$$= 10 + 2\left[\sum_{j=1}^{100} j - \sum_{j=1}^{5} j + (100-6+1)(4)\right]$$

$$= 10 + 2\left[\frac{100(101)}{2} - \frac{5(6)}{2} + 95(4)\right] = 10{,}840$$

c) The value of `time` after the following programming segment is executed

```
time = 20;
for (i = 1; i <= n; i++){
     time = time + 5
     for (j = 0; j <= n+2; j++){
              time = time + 10
     }
}
```

answer: `time` $= 10n^2 + 35n + 20$

const

$$time = 20 + \sum_{i=1}^{n}\left[5 + \sum_{j=0}^{n+2}[10]\right]$$

$$= 20 + \sum_{i=1}^{n}\left[5 + (n+2-0+1)(10)\right] = 20 + \sum_{i=1}^{n}\left[5 + 10n + 30\right]$$

$$= 20 + \sum_{i=1}^{n}\left[\underbrace{10n+35}_{\text{no } i\text{, const}}\right] = 20 + (n \cancel{-1} \cancel{+1})(10n+35) = 20 + 10n^2 + 35n$$

$$= 10n^2 + 35n + 20$$

d) The value of `time` after the following programming segment is executed

```
time = 20;
for (i = 1; i <= n; i++){
     time = time + 6i
     for (j = i; j <= n+2; j++){
               time = time + 10
     }
}
```

answer: `time` $= 8n^2 + 28n + 20$

$$\text{time} = 20 + \sum_{i=1}^{n}\left(6i + \sum_{j=i}^{n+2}(10)\right) \quad \underline{\text{OR}}$$

const

$$\text{time} = 20 + \sum_{i=1}^{n} 6i + \sum_{i=1}^{n}\sum_{j=i}^{n+2}(10)$$

$$= 20 + 6\sum_{i=1}^{n} i + \sum_{i=1}^{n}(n+2-i+1)(10)$$

Both const

$$= 20 + \overset{3}{\cancel{6}}\cdot\frac{n(n+1)}{\cancel{2}} + 10\sum_{i=1}^{n}(n+3-i)$$

$$= 20 + 3n(n+1) + 10\sum_{i=1}^{n}(n+3) - 10\sum_{i=1}^{n} i$$

$$= 20 + 3n^2 + 3n + 10(n\cancel{-1+1})(n+3) - \overset{5}{\cancel{10}}\cdot\frac{n(n+1)}{\cancel{2}}$$

$$= 20 + 3n^2 + 3n + 10n^2 + 30n - 5n^2 - 5n$$

$$= 8n^2 + 28n + 20$$

e) The value of `time` after the following programming segment is executed

```
time = 20;
for (i = 1; i <= n; i++){
     time = time + 10i
     for (j = 4; j <= n+2; j++){
             time = time + 10j
     }
}
```

$$\sum_{i=1}^{\cancel{n}\ n+2} i = \frac{n(n+1)}{2} \quad (n+2)$$

answer: `time` $= 5n^3 + 30n^2 - 25n + 20$

$$\text{time} = 20 + \sum_{i=1}^{n}\left(10i + \sum_{j=4}^{n+2} 10j\right) = 20 + \sum_{i=1}^{n}\left(10i + 10\left(\sum_{j=1}^{n+2} j - \sum_{j=1}^{3} j\right)\right)$$

Fix the bound (var)

$$= 20 + \sum_{i=1}^{n}\left(10i + \cancel{10}^{\,5}\cdot\left(\frac{(n+2)(n+3)}{\cancel{2}} - \frac{3(4)}{\cancel{2}}\right)\right)$$

$$= 20 + \sum_{i=1}^{n}\left(10i + 5\left(n^2 + 3n + 2n + 6 - 12\right)\right)$$

var, const.

$$= 20 + \sum_{i=1}^{n}\left(10i + 5\left(n^2 + 5n - 12\right)\right)$$

$$= 20 + \sum_{i=1}^{n} 10i + \sum_{i=1}^{n} 5\left(n^2 + 5n - 12\right) = 20 + 10\cdot\sum_{i=1}^{n} i + (n - \cancel{1+1})\left(5n^2 + 25n - 30\right)$$

$$= 20 + \cancel{10}^{\,5}\cdot\frac{n(n+1)}{\cancel{2}} + 5n^3 + 25n^2 - 30n = 20 + 5n^2 + 5n + 5n^3 + 25n^2 - 30n$$

$$= 5n^3 + 30n^2 - 25n + 20$$

If all of the problems can't be done in the lab period, students can work on them on their own.